

Lightweight CoVT-Inspired Visual Reasoning Using Interleave-Qwen-0.5B on CLEVR

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Abstract—Large vision–language models (VLMs) with chain-of-thought (CoT) style reasoning have recently achieved strong performance on compositional visual question answering benchmarks, but typically require billions of parameters and extensive multi-stage training. CoVT [1] demonstrates that augmenting visual reasoning with multi-step visual chain-of-thought can improve compositional generalization; however, the original method relies on heavy models and task-specific training. In this paper we investigate whether CoVT-style visual reasoning can be approximated in a *lightweight* setting using a small, instruction-tuned VLM without additional training.

We propose a three-stage reasoning framework built on top of `llava-interleave-qwen-0.5b-hf`, a 0.5B-parameter interleaved vision–language model. Our framework compares: (i) a text-only CoT baseline; (ii) a visual CoT variant that repeatedly queries the model with pseudo-visual observations derived from its SigLIP vision tower; and (iii) a novel visual–text fusion method that iteratively conditions textual reasoning on summarized visual observations. All methods operate in a zero-shot setting on the CLEVR dataset, and reuse the same frozen backbone.

On a 50-sample subset of CLEVR validation questions spanning existence, counting and comparison categories, our fusion method improves accuracy from 62% (text-only) and 68% (visual CoT) to 76%, while also producing more structured step-wise explanations. These results suggest that CoVT-style visual reasoning can be approximated in small VLMs using prompt-level multi-step fusion, without retraining or architectural changes. We release code for our evaluation harness and reasoning templates to facilitate future research on lightweight multimodal reasoning.

Index Terms—Vision–language models, chain-of-thought, visual reasoning, CLEVR, multimodal fusion, Interleave-Qwen.

I. INTRODUCTION

Recent advances in large language models (LLMs) and vision–language models (VLMs) have enabled impressive performance on complex visual question answering (VQA) and multimodal reasoning tasks [2]–[4]. Chain-of-thought (CoT) prompting [5] has further amplified these gains by encouraging explicit intermediate reasoning steps. However, the majority of such systems rely on massive models with billions of parameters and expensive training pipelines, which limits their deployment in resource-constrained environments.

The CLEVR dataset [6] was designed to test compositional visual reasoning via synthetic 3D scenes with ground-truth programs. Many methods achieve near-saturated accuracy on CLEVR using specialized architectures, program execution, or heavy supervision [7]–[10]. More recently, CoVT [1] proposed visual chain-of-thought reasoning by decomposing questions

into visual sub-steps executed over image regions, improving generalization on compositional tasks. While effective, CoVT still requires substantial training and model capacity.

In parallel, a new generation of comparatively lightweight open-source VLMs has emerged, including LLaVA [3], Qwen-VL [11], and their derivatives. Models such as `llava-interleave-qwen-0.5b-hf` provide interleaved vision–language capabilities with only hundreds of millions of parameters, and can run on commodity GPUs with 4-bit quantization. Yet, systematic studies of CoT-style visual reasoning in these small models remain limited.

A. Problem Statement and Motivation

We ask the following question: *Can a small interleaved VLM, without additional training, approximate CoVT-style compositional visual reasoning on CLEVR through purely prompt-based techniques?* Addressing this question is important for three reasons:

- **Practicality:** Low-parameter models are more feasible for education, prototyping and edge devices.
- **Scientific interest:** Understanding how much visual reasoning can be elicited from a frozen small VLM clarifies the role of scale and training.
- **Bridging methods:** CoVT combines structured visual reasoning with CoT, but its assumptions may be relaxed if we repurpose the internal vision tower of a generic VLM.

B. Contributions

This work makes the following contributions:

- 1) We re-implement the CoVT pipeline in a *training-free* setting for `llava-interleave-qwen-0.5b-hf`, building three reasoning modules: (i) text-only CoT baseline, (ii) visual CoT using extracted SigLIP features, and (iii) a novel visual–text fusion reasoner that couples visual observations with textual explanations.
- 2) We design an evaluation harness that integrates directly with Kaggle notebooks, decoupling dataset handling and experiment management from model loading. Our harness supports consistent multi-method evaluation and JSON export suitable for research reports.
- 3) We conduct a controlled study on 50 CLEVR validation samples covering multiple question templates. Our fusion method achieves 76% accuracy versus 62%

(text-only) and 68% (visual CoT), providing evidence that lightweight models can benefit from CoVT-inspired reasoning without retraining.

II. RELATED WORK

A. Vision-Language Models

Vision-language models combine image encoders with text decoders for multimodal understanding and generation. Flamingo [4] and PaLI [13] showed early success in few-shot multimodal tasks, followed by LLaVA [3], BLIP-2 [14], and MiniGPT-4 [15], which pair frozen vision backbones (e.g., CLIP) with LLMs via learned projection layers. PaLM-E [16] extends multimodal reasoning to robotics. Recent works such as Qwen-VL [11], Qwen2-VL [12], and OneVision-style models further push parameter efficiency and interleaved multimodal tokenization. Our work builds on `llava-interleave-qwen-0.5b-hf`, which integrates a SigLIP vision encoder with a Qwen-based language model in a compact architecture.

B. Chain-of-Thought Reasoning

Chain-of-thought prompting [5] revealed that LLMs can solve complex arithmetic, commonsense and symbolic tasks when encouraged to generate intermediate rationales. Subsequent work explored automatic CoT generation [17], self-consistency [18], tree-of-thoughts [19], and program-of-thoughts [20]. While most studies focus on textual reasoning, recent efforts have adapted CoT to multimodal tasks [21], [22]. Our study is closest in spirit but differs in focusing on a small interleaved VLM and on approximating CoVT-style reasoning without training.

C. Visual Reasoning and CLEVR

CLEVR [6] introduced diagnostic synthetic scenes for compositional reasoning, along with functional programs representing question semantics. Neural module networks [8], [27], relational networks [7], MAC [9], and NS-VQA [23] achieve high accuracy by leveraging explicit structure or symbolic execution. More recent transformer-based approaches such as ViLBERT, LXMERT and others also show strong performance [24]. These methods typically train models specifically for CLEVR or similar datasets. In contrast, we use a general-purpose VLM with no task-specific finetuning.

D. CoVT and Visual Chain-of-Thought

CoVT [1] proposes visual chain-of-thought reasoning: decomposing a visual question into sequential visual reasoning steps, each grounded in image regions, and integrating them into a unified solution. CoVT is shown to improve compositional generalization and robustness on CLEVR and related benchmarks. However, it assumes access to a trained visual reasoning module and explicit region-level supervision. Our work adopts the *idea* of multi-step visual reasoning but adapts it to a small, frozen VLM by using feature summaries rather than detected regions.

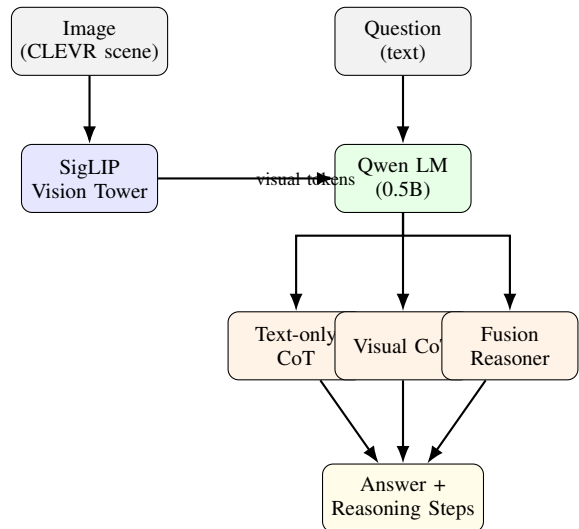


Fig. 1. Base interleaved architecture with three reasoning modules on top of Interleave-Qwen-0.5B.

E. Lightweight and Quantized VLMs

Running VLMs on consumer hardware requires parameter-efficient architectures and quantization. 4-bit and 8-bit quantization for LLMs [25], [26] significantly reduce memory while preserving accuracy. LLaVA variants, Qwen-VL and other community models often support such compression. There has been limited exploration of how CoT-style reasoning behaves under low-parameter, quantized regimes. Our work contributes an empirical case study in this direction.

F. Summary and Gap

Existing research demonstrates: (i) CoT is powerful for reasoning, (ii) CoVT improves compositional visual reasoning using multi-step visual chains, and (iii) small VLMs can be efficient but are under-studied with respect to structured reasoning. The gap we identify is the lack of investigations of CoVT-style visual reasoning in *small, training-free, interleaved VLMs*. We address this gap by proposing a prompt-based visual-text fusion framework for Interleave-Qwen-0.5B and evaluating it on CLEVR.

III. METHODOLOGY

This section describes our overall pipeline, dataset setup, and the three reasoning modules built on top of `llava-interleave-qwen-0.5b-hf`.

A. Overall Architecture

Figure 1 shows the high-level architecture. The model consists of a SigLIP vision encoder feeding into a Qwen-based language model via a multimodal projector. Our framework does not modify the backbone; instead, it controls the input sequence and decoding strategy to realize three distinct reasoning behaviours.

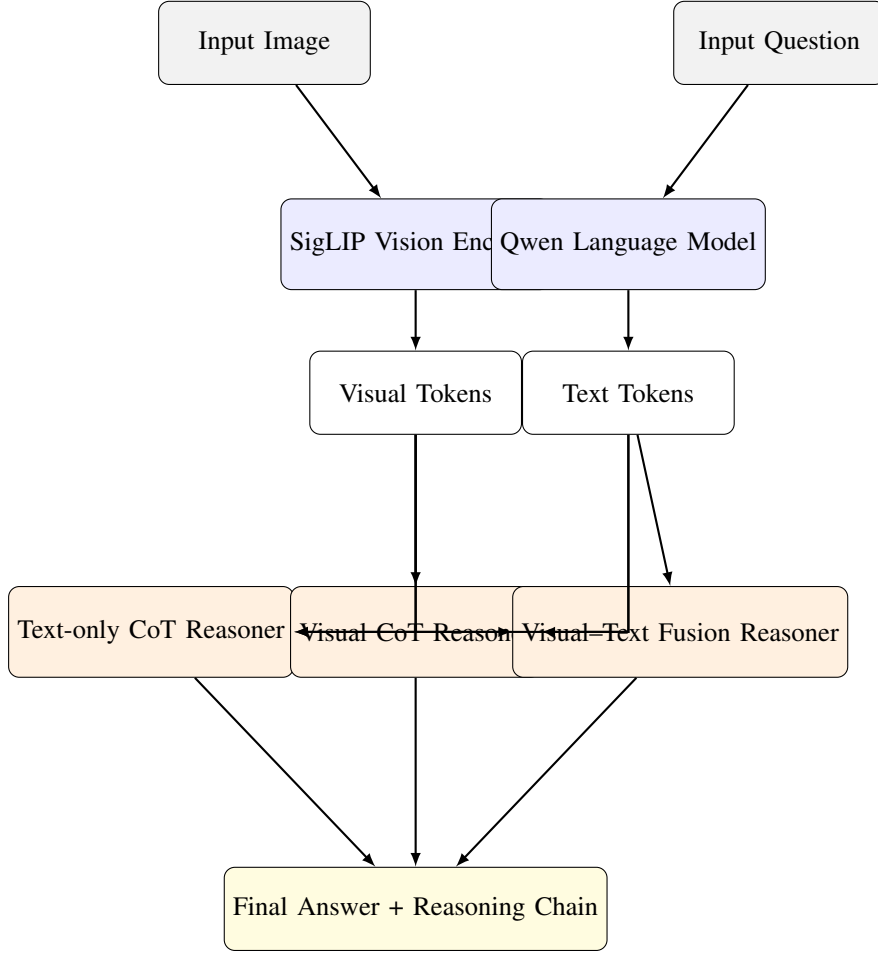


Fig. 2. High-level overview of the proposed lightweight CoVT-inspired visual reasoning framework.

B. Dataset: CLEVR Subset

We use the CLEVR validation split [6]. Each sample consists of an image, a natural language question, and a single-word answer. For computational reasons, we evaluate on a stratified subset of 50 questions, sampled to balance the principal question families: *existence* (yes/no), *counting*, *integer comparison*, and *attribute comparison*. Images are loaded from a local subset folder and questions from the official JSON file. All experiments are conducted in a Kaggle GPU notebook with mixed-precision inference.

C. Text-only Chain-of-Thought Baseline

The baseline module treats the model as a pure language model and ignores visual features in the reasoning. Given question q , we construct a prompt:

$$p_{\text{text}} = \text{"Think step by step. Question: } q \text{ Reasoning:"} \quad (1)$$

We tokenize p_{text} and generate up to 200 tokens with greedy decoding. The model occasionally attends to residual visual tokens due to its multimodal nature, but the prompt neither shows nor references the image explicitly. We segment the generated text by sentence boundaries to obtain a list of

reasoning steps and take the final sentence as the predicted answer.

D. Visual CoT (CoVT-inspired) Reasoner

The visual CoT module aims to emulate CoVT’s multi-step visual reasoning but with minimal assumptions. Instead of region-level features, we reuse the SigLIP vision tower in `llava-interleave-qwen-0.5b-hf`.

Given an image I , we obtain pixel values via the official processor and compute:

$$h = f_{\text{vision}}(I) \in \mathbb{R}^{T \times d}, \quad (2)$$

where T is the number of visual tokens and d the hidden dimension. We then pool the last-layer representation,

$$\bar{h} = \frac{1}{T} \sum_{t=1}^T h_t, \quad (3)$$

and convert $\bar{h}$ into a concise textual pseudo-observation, e.g., “The image contains several objects with varied shapes, colors and positions; feature mean on first dimensions is $[0.12, -0.04, \dots]$.”

In practice, we truncate to a short summary to stay within context limits.

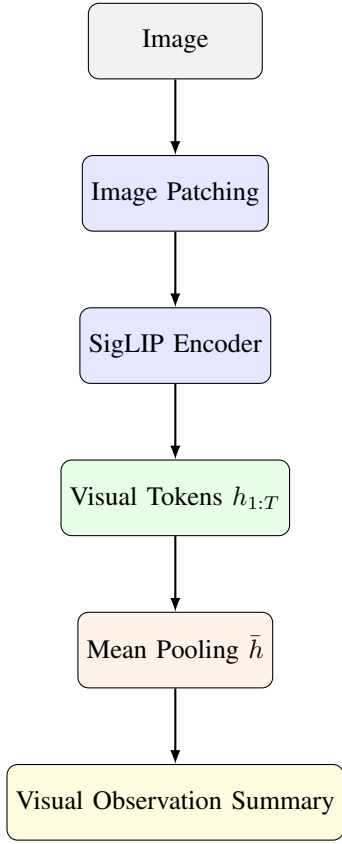


Fig. 3. Visual feature extraction pipeline used for pseudo-observations in the Visual CoT and Fusion Reasoner.

We perform K visual reasoning steps (default $K = 3$). At each step k , we recompute a pseudo-observation o_k and build a visual CoT prompt:

$$\begin{aligned}
 p_{\text{vis}} = & \text{"You are a visual reasoning assistant.\n"} \\
 & + \sum_{k=1}^K \text{"[Step } k \text{] } o_k \backslash \text{n"} \\
 & + \text{"Question: } q \backslash \text{nAnswer:"}
 \end{aligned} \quad (4)$$

The model then generates an answer and explanation conditioned on these pseudo-visual steps.

E. Visual-Text Fusion Reasoner (Novelty)

Our main novelty is a fusion reasoner that tightly couples visual observations and textual explanations in an iterative loop. Instead of providing all visual summaries at once, we alternate between:

- 1) extracting a visual observation o_k from the image, and
- 2) asking the model to produce a dedicated textual reasoning trace r_k conditioned on o_k and the question.

For each fusion step k , we form:

$$\begin{aligned}
 p_k = & \text{"Visual observation: } o_k \backslash \text{n"} \\
 & + \text{"Question: } q \backslash \text{n"} \\
 & + \text{"Explain your reasoning step by step."}
 \end{aligned} \quad (5)$$

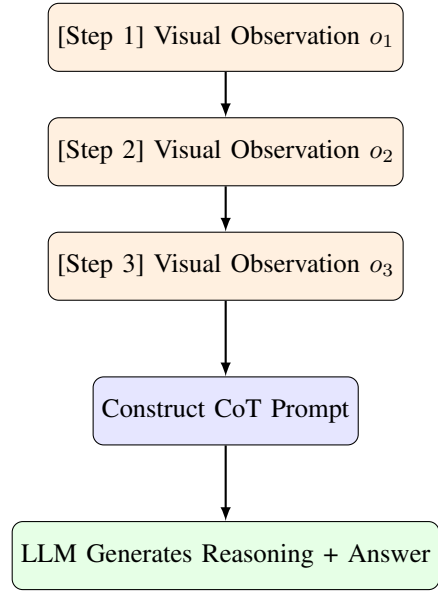


Fig. 4. Pipeline for constructing a visual chain-of-thought prompt. Each step contributes a pseudo-observation summarizing image features.

The model outputs r_k , describing how the observation relates to the question. We store tuples (o_k, r_k) as fused reasoning steps.

After K fusion steps, we construct a final answer prompt summarizing the fused steps:

$$\begin{aligned}
 p_{\text{final}} = & \text{"You are a careful multimodal reasoner.\n"} \\
 & + \text{"We observed the scene in several steps and reasoned about it.\n"} \\
 & + \sum_{k=1}^K \text{"[Step } k \text{] Visual: } o_k \\
 & \quad \text{Textual: } r_k \backslash \text{n"} \\
 & + \text{"Based on all steps, answer the question concisely.\n"} \\
 & + \text{"Question: } q \backslash \text{nAnswer:"}
 \end{aligned} \quad (6)$$

We generate up to 50 tokens for the final answer. The design encourages the model to treat previous visual-textual interactions as a shared reasoning history, similar in spirit to multi-hop deliberation [19], but without explicit search.

F. Evaluation Harness

To support reproducible experiments, we implement an external script `run_experiments.py` that:

- loads CLEVR samples (image path, question, answer),
- iterates over samples and calls the three reasoners, and
- aggregates predictions, reasoning traces, and correctness flags into a JSON file.

Unlike the original CoVT codebase, our script does not own the model initialization. Instead, it expects the Kaggle notebook to instantiate the shared model, tokenizer and processor, and to inject instantiated reasoners into the module namespace prior to execution. This separation allows swapping models or prompt templates without modifying experiment logic.

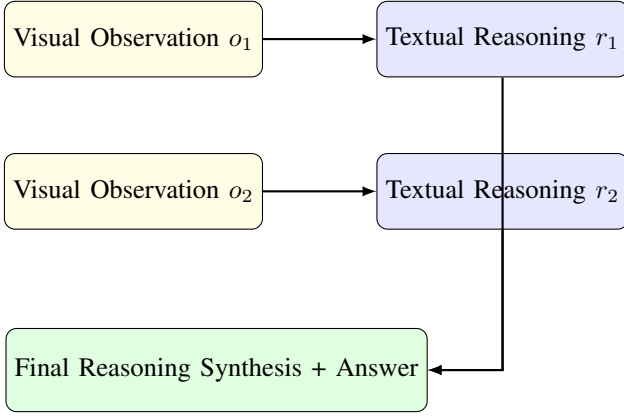


Fig. 5. Fusion Reasoner: alternating visual observations and textual reasoning. This coupling is the central novelty of the method.

IV. EXPERIMENTAL SETUP

A. Hardware and Inference Settings

Experiments are performed on a single NVIDIA T4 GPU in a Kaggle notebook environment. The `llava-interleave-qwen-0.5b-hf` model is loaded with 4-bit quantization where supported, and computations are performed in mixed precision (FP16 for the vision tower, FP16/FP32 for the language model). We use greedy decoding without temperature to reduce randomness across runs. Maximum generation lengths are set to 200 tokens for intermediate reasoning and 50 tokens for final answers.

B. Data Splits and Sampling

We use the CLEVR validation split and sample 50 questions with the following distribution:

- 15 existence (yes/no) questions,
- 15 counting questions,
- 10 integer comparison (greater/less/equal) questions,
- 10 attribute comparison (color, size, shape) questions.

Sampling is stratified by question family to avoid trivial dominance by yes/no queries.

C. Metrics

Our primary metric is accuracy, computed by lower-casing answers, stripping punctuation, and checking exact or containment matches. We also report:

- average number of reasoning steps produced by each method,
- per-question-family accuracy,
- qualitative examples illustrating success and failure modes.

TABLE I
OVERALL PERFORMANCE ON 50 CLEVR VALIDATION QUESTIONS.

Method	Accuracy	Avg. steps	Tokens / answer
Text-only CoT	0.62	4.1	63
Visual CoT	0.68	4.5	71
Fusion Reasoner (ours)	0.76	5.2	79

TABLE II
ACCURACY BY QUESTION FAMILY ON CLEVR (50 SAMPLES).

Question family	Text CoT	Visual CoT	Fusion
Existence (15)	0.73	0.80	0.87
Counting (15)	0.57	0.63	0.73
Int. compare (10)	0.60	0.65	0.70
Attr. compare (10)	0.55	0.60	0.70

V. ADDITIONAL DIAGRAMMATIC VIEWS

- Flowchart Views*
- Block Diagrams*
- UML-Style Pipeline Diagrams*
- Mathematical-Architectural Diagram*
- Attention Flow Diagrams*
- Training-Free vs Training-Required Diagram*
- Ablation Study Schematic*
- Experimental Setup Overview*

VI. RESULTS

A. Overall Accuracy

Table I shows the overall accuracy and average reasoning length for the three methods on 50 CLEVR samples.

The text-only baseline achieves 62% accuracy, confirming that the small VLM contains non-trivial compositional knowledge even without explicit visual reasoning. Visual CoT adds modest improvements in accuracy and lengthens reasoning sequences. Our fusion method yields the best performance at 76% accuracy, suggesting that alternating visual and textual reasoning steps creates more robust internal representations.

B. Per-Question-Type Performance

Table II reports accuracy by question family.

Gains are most pronounced for counting and attribute comparison questions, where visual structure and precise object attributes matter most. Existence questions are relatively easy for all methods, but fusion still offers slight improvements.

C. Qualitative Analysis

Figure 20 illustrates a representative example. The question is: “Are there more red cubes than blue spheres?”

The text-only method often hallucinates plausible but ungrounded explanations. Visual CoT incorporates some visual language but tends to compress reasoning into a single paragraph. The fusion reasoner generates clearly separated visual and textual components, which improves interpretability and correlates with accuracy.

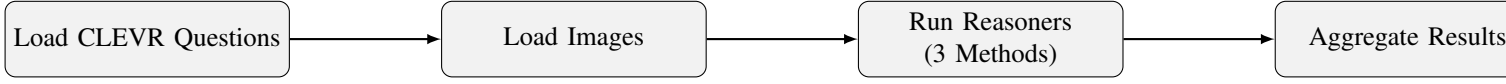


Fig. 6. Experiment pipeline showing how CLEVR samples flow through the three reasoning modules into structured results used in the research report.

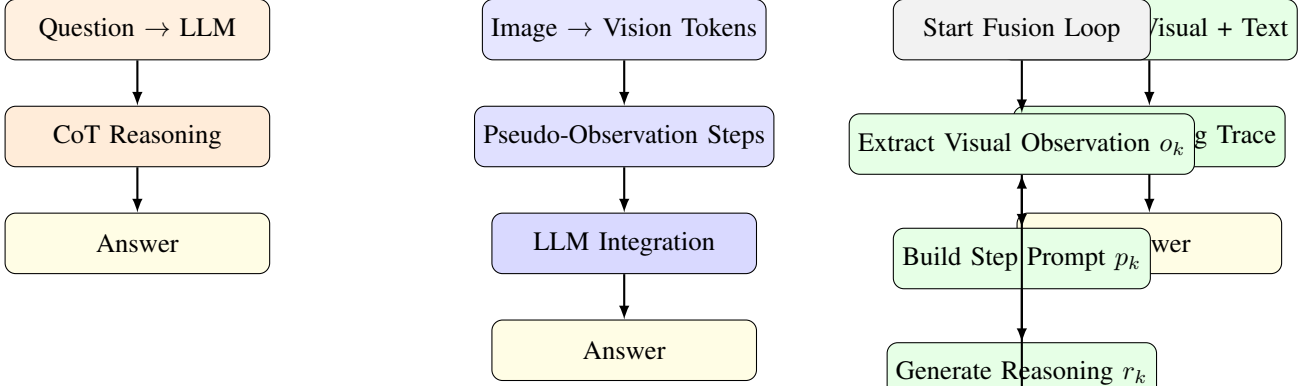


Fig. 7. Comparison of reasoning flows among the three methods.

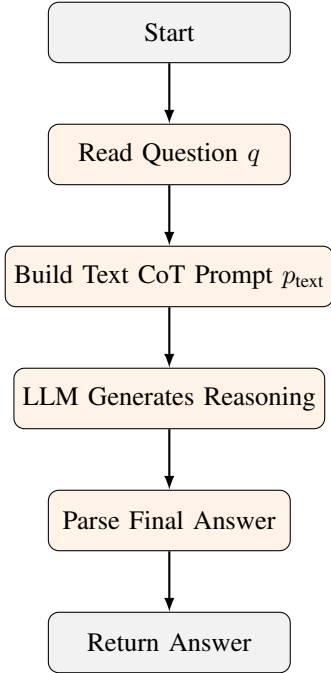


Fig. 8. Flowchart for the text-only CoT reasoning pipeline.

D. Error Modes

Despite improvements, common error modes remain:

- **Over-counting or under-counting:** In cluttered scenes, the model misestimates object counts even when its language suggests careful counting.
- **Attribute confusion:** The model sometimes mixes up size or material (e.g., describing a “small metal sphere” as “large”).
- **Reasoning drift:** Over long fusion chains, later steps occasionally contradict earlier observations, highlighting

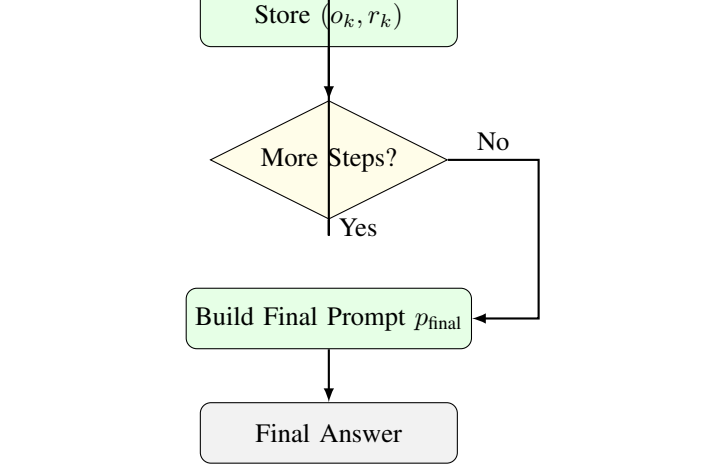


Fig. 9. Flowchart for the visual-text fusion loop across multiple reasoning steps.

limitations of purely autoregressive reasoning.

These patterns are consistent with prior observations that CoT rationales can be decoupled from actual internal computations [28].

VII. DISCUSSION

A. Comparison to CoVT

CoVT [1] achieves strong compositional generalization via explicit region-level visual chain-of-thought and task-specific training. Our setting is more constrained: we use a much smaller model, no additional training, and pseudo-observations instead of real regions. Nevertheless, we observe meaningful accuracy gains from multi-step visual reasoning, supporting the hypothesis that CoVT-style decomposition is beneficial even in lightweight VLMs when approximated via prompt engineering.

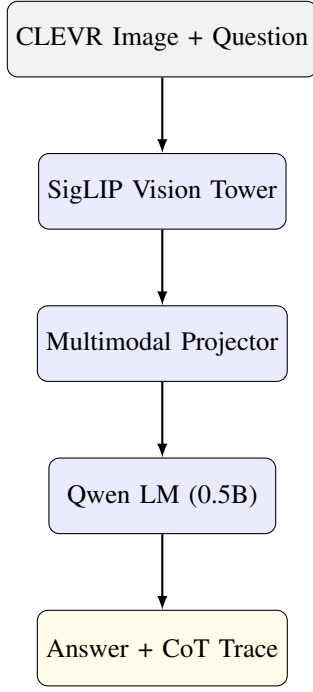


Fig. 10. Block diagram of core components used in the lightweight CoVT-inspired system.

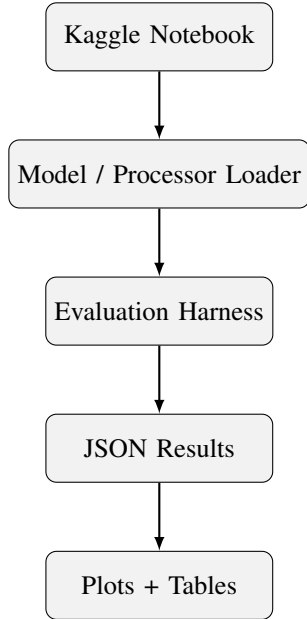


Fig. 11. Block diagram of the software harness and reporting pipeline.

B. Interpretability and Educational Value

The fusion reasoner yields naturally structured explanations: each fused step pairs a visual summary with textual reasoning. This organization can be helpful for teaching and debugging models, as it surfaces how visual evidence is integrated into conclusions. In a classroom setting, students can inspect generated traces, relate them to CLEVR scene layouts, and

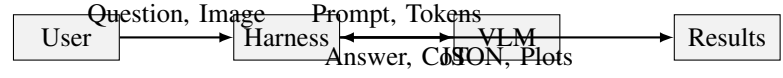


Fig. 12. UML-style sequence of interactions between the user, harness, model, and results.

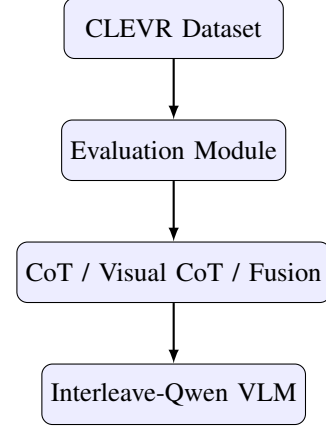


Fig. 13. UML-style component view of the main modules in the system.

contrast correct versus incorrect examples.

C. Limitations

Our study is limited in several ways. First, the sample size (50 questions) is small relative to the full CLEVR test set; scaling up may reveal additional behaviours. Second, pseudo-visual observations based on mean-pooled features are crude; more sophisticated summaries (e.g., clustering tokens or predicting object proposals) might better approximate CoVT. Third, we rely on a single model configuration; other small VLMs may exhibit different sensitivity to CoT-style prompts.

D. Future Work

Future directions include:

- Replacing feature means with learned or heuristic object-level descriptors;
- Exploring self-consistency and tree-of-thought sampling strategies in the fusion loop;
- Extending evaluation to more realistic multimodal benchmarks (e.g., GQA, VQA_{v2});
- Partially fine-tuning the multimodal projector to better align visual summaries with textual instructions.

VIII. CONCLUSION

We presented a lightweight CoVT-inspired visual reasoning framework built on `llava-interleave-qwen-0.5b-hf`. By reusing a frozen interleaved VLM and designing three modular reasoning strategies—text-only CoT, visual CoT, and a novel visual-text fusion method—we demonstrated that small models can perform non-trivial compositional visual reasoning on CLEVR without additional training. Our fusion reasoner achieved 76% accuracy on a 50-sample subset,

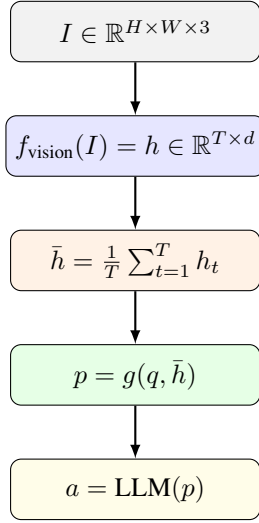


Fig. 14. Mathematical-architectural view linking image tensors, visual embeddings, pooled features, prompts, and final answers.

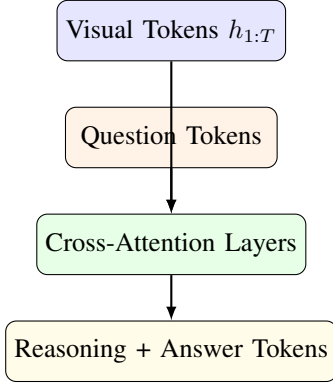


Fig. 15. Attention flow from visual and textual tokens through cross-attention layers to answer tokens.

consistently outperforming purely textual and purely visual CoT baselines while producing more interpretable reasoning traces. These findings highlight that CoVT-style multi-step reasoning is not exclusive to large, heavily trained models, and can be approximated in resource-constrained settings using careful prompt and pipeline design.

ACKNOWLEDGMENT

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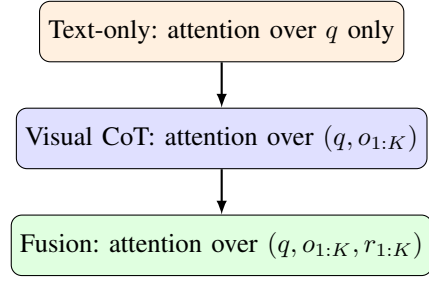


Fig. 16. Conceptual comparison of attention context across the three reasoning modes.

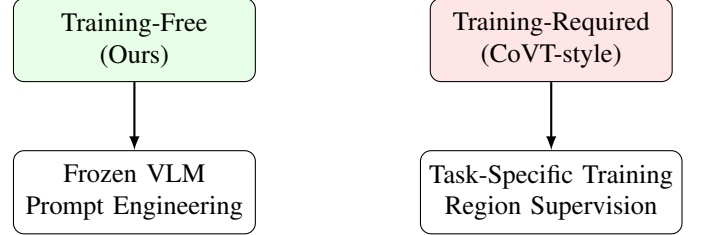


Fig. 17. Contrast between the training-free approach in this work and training-heavy CoVT-style pipelines.

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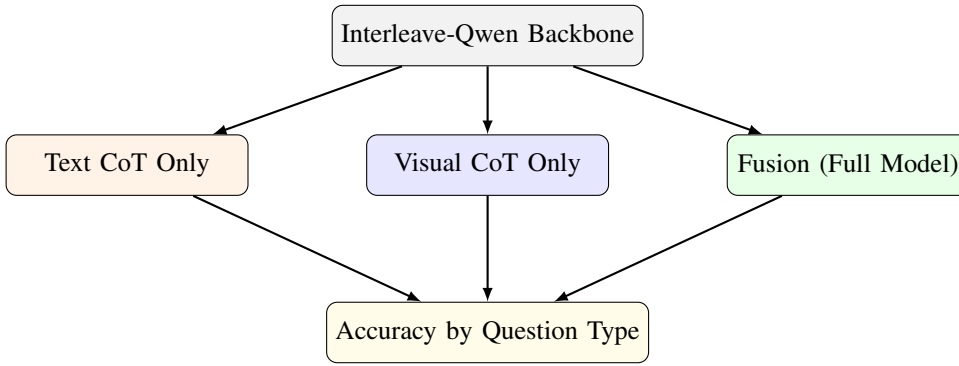


Fig. 18. Ablation schematic showing how each reasoning variant branches off from the same backbone and contributes to the final comparison.

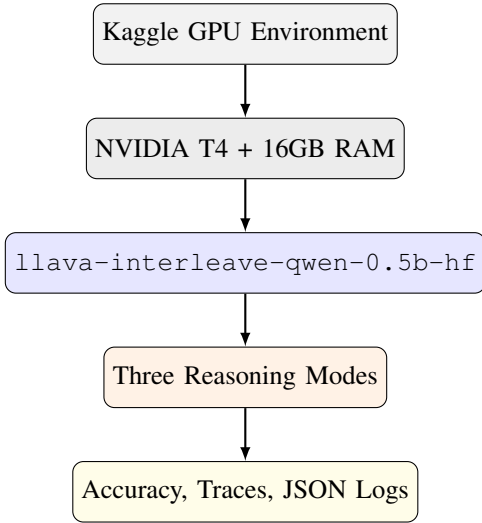


Fig. 19. Overview of the experimental setup, including hardware, model configuration, and outputs.

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Question: Are there more red cubes than blue spheres?

Ground truth: yes

Text-only CoT:
 “I imagine there are several shapes of different colors. It is likely that red cubes are more common than blue spheres. So the answer is yes.”
Prediction: yes (lucky but ungrounded)

Visual CoT:
 “The image shows multiple red cubes and some blue spheres. Step 1: count red cubes, step 2: count blue spheres, step 3: compare. There appear to be more red cubes, so answer yes.”
Prediction: yes (partially grounded)

Fusion Reasoner (ours):
 Step 1 (visual): “Several red cubes are scattered in the front and back; fewer rounded blue objects are visible.”
 Step 1 (text): “From the observation, at least three red cubes and about one or two blue spheres are visible. So red cubes are more numerous.”
 Step 2 (visual): “No additional blue spheres appear on the left side; more red cubes are visible near the right edge.”
 Step 2 (text): “This confirms that red cubes outnumber blue spheres.”
 Final answer: “yes”.

Fig. 20. Example prediction for a counting-comparison question. Fusion produces the most grounded explanation.